

Linear Algebra – MAT 2610

Section 2.2 (The Inverse of a Matrix)

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Matrix Inverse

Definition: An $n \times n$ matrix A is **invertible** if there exists an $n \times n$ matrix B such that $\underline{AB} = \underline{BA} = \underline{I_n}$. We say B is the **inverse** of A , denoted by $\underline{A^{-1}}$.

Example: If $\underline{A} = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$ and $\underline{B} = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$ then $AB = BA = I_2$

A matrix that is not invertible is called a **singular** matrix. A matrix that is invertible is called a **non-singular** matrix.

Theorem 4

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ with $\underline{ad - bc} \neq 0$ then

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

The value $\underline{ad - bc}$ is called the **determinant** of A and is denoted $\underline{\det(A)}$



Theorem 5

If A is an invertible $n \times n$ matrix, then for each b in R^n , the equation $Ax = b$ has the unique solution $x = A^{-1}b$



Suppose A is invertible and we know A^{-1} . Then we can use this to solve $Ax = b$

$$Ax = b$$

$$A^{-1}(Ax) = A^{-1}b \quad \leftarrow$$

$$\underline{(A^{-1}A)}x = A^{-1}b \quad \leftarrow$$

$$\underline{I_n}x = A^{-1}b \quad \leftarrow$$

$$x = \underline{A^{-1}b} \quad \leftarrow$$



Example: We know that if $A = \begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$ then $A^{-1} = \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$

Solve:

$$\begin{aligned}x_1 + 2x_2 &= 4 \\ 3x_1 + 5x_2 &= 3\end{aligned}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \quad \text{so}$$

$$\begin{aligned}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} &= \begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} -20 + 6 \\ 12 - 3 \end{bmatrix} = \begin{bmatrix} -14 \\ 9 \end{bmatrix}\end{aligned}$$



Theorem 6

If A is an invertible matrix, then A^{-1} is invertible and $(A^{-1})^{-1} = A$

If A and B are $n \times n$ invertible matrices, then so is AB and $(AB)^{-1} = B^{-1}A^{-1}$

If A is an invertible matrix, then so is A^T and $(A^T)^{-1} = (A^{-1})^T$



Definition: An **Elementary Matrix** is a matrix obtained from performing a single Elementary Row Operation on an Identity Matrix.

Examples:

$$\text{\textit{I}}_4 \xrightarrow{sr_2 \rightarrow r_2} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & s & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{\textit{I}}_4 \xrightarrow{kr_1 + r_3 \rightarrow r_3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ k & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Theorem 7

An $n \times n$ matrix A is invertible if and only if A is row equivalent to I_n . The sequence of row operations that reduces A to I_n also transforms I_n to A^{-1}



Example: If $A = \begin{bmatrix} 2 & -3 \\ 3 & -5 \end{bmatrix}$, find A^{-1} if it exists

$$\begin{aligned} \begin{bmatrix} 2 & -3 \\ 3 & -5 \end{bmatrix} &\xrightarrow{\frac{1}{2}r_1 \rightarrow r_1} \begin{bmatrix} 1 & -3/2 \\ 3 & -5 \end{bmatrix} \xrightarrow{r_2 - 3r_1 \rightarrow r_2} \begin{bmatrix} 1 & -3/2 \\ 0 & -1/2 \end{bmatrix} \\ &\xrightarrow{-2r_2 \rightarrow r_2} \begin{bmatrix} 1 & -3/2 \\ 0 & 1 \end{bmatrix} \xrightarrow{r_1 + \frac{3}{2}r_2 \rightarrow r_1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} &\xrightarrow{\frac{1}{2}r_1 \rightarrow r_1} \begin{bmatrix} 1/2 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{r_2 - 3r_1 \rightarrow r_2} \begin{bmatrix} 1/2 & 0 \\ -3/2 & 1 \end{bmatrix} \\ &\xrightarrow{-2r_2 \rightarrow r_2} \begin{bmatrix} 1/2 & 0 \\ 3 & -2 \end{bmatrix} \xrightarrow{r_1 + \frac{3}{2}r_2 \rightarrow r_1} \begin{bmatrix} 5 & -3 \\ 3 & -2 \end{bmatrix} = A^{-1} \end{aligned}$$



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